

## Professional Summary

*“When an implementation hits a wall, my instinct is to look for the mathematical structure that removes it – not the hardware that pushes through it.”*

Versatile R&D Engineer (**16 years of hands-on R&D experience**) with a unique background bridging low-level embedded systems, power electronics and cybernetics with modern enterprise architectures (FinTech) across Java, .NET and React ecosystems. Core expertise: **mathematical complexity reduction** – replacing costly infrastructure and hardware constraints with clever deterministic models (OpEx  $\rightarrow$  0). Commercial product author and creator of proprietary AI/ML frameworks (FochBot, ARR).

## Skills Matrix & Technologies

|                                        |                                                                                                                    |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Software Engineering</b>            | Java, C# (.NET), React, TypeScript, Python, FastAPI, Docker, Spring Boot                                           |
| <b>AI / ML (upper layers)</b>          | ChromaDB, LangChain, multi-LLM routing, cognitive architectures with persistent memory (ARR)                       |
| <b>Embedded, PCB &amp; IoT</b>         | C / ARM Assembly (Cortex M0/M4F), Q31 fixed-point, AVR / ATmega / ATtiny, KiCad, Altium, Home Assistant, MQTT, FSM |
| <b>R&amp;D Math &amp; Data Science</b> | Number theory, NumPy, Numba JIT, Haskell, Lean 4, trajectory modelling, telemetry systems                          |
| <b>Management &amp; Cybernetics</b>    | Engineering office coordination, service management, triac phase control, CNC systems (CO <sub>2</sub> laser)      |

## Work Experience

### Commerzbank AG

2019 – present

*Software Developer (Java / React / C#)*

- Design, development and maintenance of backend systems in a rigorous FinTech environment.
- Technology evolution within the bank: C# .NET  $\rightarrow$  Frontend (React / TypeScript)  $\rightarrow$  current specialisation as backend Java developer.
- Current project: business-banking platform (10+ microservices, Java / Maven / Spring Boot, Oracle DB, Kafka); responsible for the full backend layer, including platform-wide infrastructure improvements (e.g. shared logback configuration service across all microservices); CI/CD: TeamCity  $\rightarrow$  ArgoCD  $\rightarrow$  OpenShift.

### Owner Delfi Electronics Sp. z o.o.

2013 – present

*R&D Engineer / CNC Systems Architect*

- Designed and commissioned an automated wood-processing production line based on a modified 50 W CO<sub>2</sub> laser (custom air-assist, chiller, Ruida DSP controller calibration).
- Production deployment and commercialisation of over 1 000 finished products. Link to site: Delfi Laser
- End-to-end custom electronic device and control algorithm design (full cycle: schematic  $\rightarrow$  PCB  $\rightarrow$  firmware).
- **Project #01 – Autonomous Boiler Controller (2016):**  $\pm 0.1$  °C on a coal boiler; 5.5 years production;  $\sim 11$ M records; basis for 3 engineering theses (5/5).
- **Project #03 – LPG Injector Calibrator:** Scatter  $> 0.3$  mm  $\rightarrow$   $< 0.01$  mm; event-loop feedback replaced caliper measurement; homologation approved.
- **Project #04 – WST/WSTK Automotive Voltmeter (2011–2012):** Engine-state FSM voltmeter; **150 units** (car) + 15 units (motorcycle) sold nationwide. [Demo](#)
- **Project #11 – AutoMobilClub Łódź Timing System (2011):** Commercial timing system (lap 0.01 s, reaction 0.001 s); predecessor of DELFI Racing RT1/RT2. [Press coverage](#)

**Engineering Office Coordinator**

2017 – 2019

- Promoted to Coordinator within 6 months of joining the R&D department after completing the engineering degree.
- Managed a team of designers, electronics engineers and embedded programmers.
- Oversaw R&D processes, new product introduction (prototype → mass production) and BOM cost optimisation.
- **Key R&D Achievement – Project LX-DK:** Solved an MCU flash-memory shortage preventing a new password-reset feature from being added to firmware. Instead of modifying the firmware, leveraged the existing UART protocol and built an independent hardware module one architectural layer above. **Result:** 500+ units sold to the manufacturer over 7 years (including 3 years after leaving the company).

**Head of Service Department**

2013 – 2017

- Promoted to Head of Department after just 6 months due to radical operational improvements.
- Cost & Process Optimisation: Restructured the entire service division; designed and implemented a comprehensive “Service Book” with standardised fault categorisation and error-coding system for automated failure-rate reporting.
- Financial Impact: Standardised pricing and raised repair quality control, successfully reducing annual service maintenance costs by over **50,000 PLN/year**.

**APH Elektra**

2009 – 2011

*Technical Advisor / Component Specialist*

- Technical consulting and deep component market analysis (equivalents, substitutes) – foundation for mass-production BOM cost optimisation.

**Key R&D Projects**

*Full technical case studies at [flamandzkiartur.com](http://flamandzkiartur.com) · R&D Portfolio PDF available on request.*

**FochBot – Cognitive Middleware for AI**

Python FastAPI ChromaDB Docker

2026–

- Two-layer cognitive middleware: persistent memory map (ARR-indexed ChromaDB) + autonomous controller that routes queries and eliminates re-derivation of already known context — token cost = 0 for local tasks.

**TrueSynth / Prime Engine**

C ARM Assembly STM32 M0/M4F Q31 Fixed-Point

2025–2026

- Redefined angle as prime fraction ( $\Delta = k/p$ ): every point lies algebraically on the unit circle — eliminates interpolation error and timing jitter at the source.
- **Result:** ppb = 0, 3 cycles/pair on Cortex-M4F, **4.7× faster** than CMSIS-DSP Q31, no FPU, zero drift after  $10^9$  steps.

**ARR Framework (Relational Rhythm Analyser) & Relational Mathematics**

- Formalised a proprietary decision-making framework based on symbolic trend analysis ( $\uparrow / \rightarrow / \downarrow$ ) instead of absolute values – designed for Edge/RT devices with extremely constrained resources.
- Published **6 scientific preprints** (Zenodo) including a hyperbolic proof of the Bertrand–Chebyshev theorem and a rotational model of prime numbers.

I consent to the processing of my personal data for the purposes necessary to carry out the recruitment process in accordance with Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 (GDPR).